



LINEAR ALGEBRA AND ITS APPLICATIONS

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CLASS-7

PROBLEMS ON EIGEN VALUES AND EIGEN VECTORS AND CAYLEY-HAMILTON THEOREM

Problem:-

1. Find the Eigen values & Eigen vectors for the given

matrix $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$

Ans: Characteristic eqn is $|A - \lambda I| = 0$

$$\begin{vmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{vmatrix} = 0 \Rightarrow (6-\lambda)[(3-\lambda)^2 - 1] + 2[-2(3-\lambda) + 2] + 2(2 - 2(3-\lambda)) = 0$$

$$\Rightarrow \lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$$

$\lambda = 8$ is one of the roots.

$$\begin{array}{r|rrrr} 8 & 1 & -12 & 36 & -32 \\ & 0 & 8 & -32 & 32 \\ \hline & 1 & -4 & 4 & 0 \end{array}$$

$$\Rightarrow \lambda^2 - 4\lambda + 4 = 0$$

$$\Rightarrow \lambda = 2, 2$$

$$\therefore \lambda_1 = 8, \lambda_2 = 2, \lambda_3 = 2$$

Consider $(A - \lambda I)x = 0$

$$\Rightarrow (6 - \lambda)x - 2y + 2z = 0$$

$$-2x + (3 - \lambda)y - z = 0$$

$$2x - y + (3 - \lambda)z = 0$$

When $\lambda = 8$,

$$\left. \begin{aligned} -2x - 2y + 2z &= 0 \\ -2x - 5y - z &= 0 \\ 2x - y - 5z &= 0 \end{aligned} \right\}$$

On solving $x_1 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$

When $\lambda = 2$,

$$\left. \begin{aligned} 4x - 2y + 2z &= 0 \\ -2x + 4y - z &= 0 \\ 2x - y + z &= 0 \end{aligned} \right\}$$

On solving $x_2 = \begin{bmatrix} \frac{k_1 - k_2}{2} \\ k_1 \\ k_2 \end{bmatrix} = x_3$

Where k_1 & k_2 are arbitrary & can take any value.

$\therefore$ Eigen vector is $x = [x_1, x_2, x_3]$

2. Find the Eigen values & Eigen vectors for the matrix

$$A = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ -1 & 1 & -1 \end{bmatrix}$$

Ans: The characteristic equation is $|A - \lambda I| = 0$

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & -1 & 1 \\ 1 & -\lambda & 0 \\ -1 & 1 & -1-\lambda \end{vmatrix} = 0 \Rightarrow \lambda^3 + \lambda = 0$$
$$\Rightarrow \lambda = 0, \lambda^2 + 1 = 0$$
$$\lambda = 0, \pm i$$

$\therefore \lambda_1 = 0, \lambda_2 = i, \lambda_3 = -i$ are the required eigen values.

Now consider $(A - \lambda I)x = 0 \Rightarrow$

$$\begin{aligned} (1-\lambda)x - y + z &= 0 \\ x - \lambda y &= 0 \\ -x + y + (-1-\lambda)z &= 0 \end{aligned}$$

Case 1:- When $\lambda = 0$

$$x - y + z = 0$$

$$x = 0$$

$$-x + y - z = 0$$

} on putting $x_1 = \begin{bmatrix} 0 \\ k_1 \\ k_2 \end{bmatrix}$, let $k_1 = k_2 = 1$

$$\therefore x_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Case 2:- When $\lambda = i \Rightarrow (1-i)x - y + z = 0$
 $x - iy + 0 \cdot z = 0$
 $-x + y + (-1-i)z = 0$

$$x_2 = \begin{bmatrix} i \\ 1 \\ -i \end{bmatrix}$$

Case 3:- When $\lambda = -i$,

$$\left. \begin{aligned} (1+i)x - y + z &= 0 \\ x + iy + 0z &= 0 \\ -x + y + (-1+i)z &= 0 \end{aligned} \right\} x_3 = \begin{bmatrix} -i \\ 1 \\ i \end{bmatrix}$$

$\therefore$ Eigen vectors corresponding to $\lambda = 0, i, -i$ is

$$X = [x_1 \ x_2 \ x_3] = \begin{bmatrix} 0 & i & -i \\ 1 & 1 & 1 \\ 1 & -i & i \end{bmatrix}$$

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Problem:- If $A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$. Evaluate A^{-1} & A^{-2}
using Cayley-Hamilton theorem of (C-H) theorem.

Ans:- $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 4-\lambda & 6 & 6 \\ 1 & 3-\lambda & 2 \\ -1 & -4 & -3-\lambda \end{vmatrix} = 0 \Rightarrow \lambda^3 - 4\lambda^2 - \lambda + 4 = 0$$

According to C-H theorem $(\lambda \rightarrow A)$

$$A^3 - 4A^2 - A + 4I = 0 \rightarrow (1)$$

Now to find A^{-1} & A^{-2}

To find A^{-1}

Multiplying eqn (1) by A^{-1} ,

$$\Rightarrow A^2 - 4A - I + 4A^{-1} = 0$$

$$\Rightarrow A^{-1} = \frac{1}{4} [-A^2 + 4A + I]$$

$$A^{-1} = \frac{1}{4} \left[- \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix} \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix} + 4 \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right]$$

$$\therefore A^{-1} = \frac{1}{4} \begin{bmatrix} 1 & 6 & 6 \\ -1 & 6 & 2 \\ 1 & -10 & -6 \end{bmatrix}$$

To find A^{-2} :

Multiplying eqn (i) by A^{-2}

$$A - 4I - A^{-1} + 4A^{-2} = 0$$

$$\Rightarrow A^{-2} = \frac{1}{4} [-A + A^{-1} + 4I]$$

$$A^{-2} = \frac{1}{4} \begin{bmatrix} 1 & -18 & -18 \\ -5 & 10 & -6 \\ 5 & 6 & 22 \end{bmatrix}$$



THANK YOU

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